Feature-Level vs Sample-Level Weights

Category-shift cold-start on text multiclass
(FULL LIVE holdout; no EWMA)

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Abstract

Both weight channels share one FSDS RF-domain attribution (LIVE vs REF ⇒ VIMP), then act on *orthogonal axes*: **feature-level** rescales coordinates ($x_{ij} \leftarrow x_{ij}\, w_j^{\text{feat}}$); **sample-level** rescales CE mass (`partial_fit(..., sample_weight=`w_i`)`). **Cold-start gate:** once a novel category reaches $n_c = 200$ samples, stop empowering it ($w_i \leftarrow 1$; feature scale off when no cold-start mass remains). Under the gate, early-n lift is clear; past the cap schemes stay near uniform (no over-weighting).

1 Shared upstream: RF-domain VIMP

Let $X^{\text{ref}} \in \mathbb{R}^{n_0 \times p}$ be the known-category reference and $X^{\text{live}} \in \mathbb{R}^{n_1 \times p}$ the live batch (known + novel). Fit a RF on

$$W_i = \begin{cases} 0 & i \in \text{REF}, \\ 1 & i \in \text{LIVE}, \end{cases}$$

and take feature importances

$$v \in \Delta^{p-1}, \qquad v_j = \frac{\text{Imp}_j}{\sum_k \text{Imp}_k}.$$

Coordinates with large v_j are those that best separate the live batch from the reference—the domain-shift footprint under category arrival.

2 Feature-level weights (geometry)

Normalize relative to the top attributed coordinate and stretch:

$$\tilde{v}_j = \frac{v_j}{\max_k v_k}, \qquad w_j^{\text{feat}} = 1 + \alpha\, \tilde{v}_j \qquad (\alpha = 3 \text{ default}). \tag{1}$$

Apply the same scale to REF and LIVE,

$$x_{ij} \;\leftarrow\; x_{ij}\, w_j^{\text{feat}}, \tag{2}$$

then re-init the CE head on scaled REF so coefficients live in the new geometry. Top VIMP coords get $\approx 1 + \alpha = 4\times$; quiet coords stay near 1.

Role. Puts CE gradient budget on the axes that carry the category shift. Especially natural for TF-IDF / BoW: a few token dimensions encode the new category; most coordinates are noise. Feature-level does *not* drop samples—it reweights the representation.

3 Sample-level weights (mass)

3.1 Membership in the attributed subspace

Take the top-K VIMP indices $\mathcal{S} = \arg\operatorname{top}_K(v)$. On $\mathcal{S}$, z-score LIVE against REF mean/std and set

$$d_i = \|z_i^{(\mathcal{S})}\|_2, \qquad m_i = \operatorname{clip}\Big(\frac{d_i - q_{0.2}(d)}{q_{0.8}(d) - q_{0.2}(d)}, 0, 1\Big). \tag{3}$$

High m_i = far from REF in the shift subspace = "new-category-like". Optional private-live fuse (labels available at adapt time):

$$m_i \;\leftarrow\; \max\big(m_i,\, \mathbf{1}[y_i \in \mathcal{C}_{\text{novel}}]\big). \tag{4}$$

3.2 Soft + hard → fused

$$w_i^{\text{soft}} = 1 + \lambda_{\text{eff}}\, m_i, \tag{5}$$

$$w_i^{\text{subset}} = \begin{cases} \beta_{\text{eff}} & m_i \geq q, \\ 1 & \text{else}, \end{cases} \tag{6}$$

$$w_i = \frac{n\,(w^{\text{soft}} \odot w^{\text{subset}})_i}{\sum_{i'}(w^{\text{soft}} \odot w^{\text{subset}})_{i'}}. \tag{7}$$

$\lambda_{\text{eff}}, \beta_{\text{eff}}$ grow with membership concentration (sharp footprint ⇒ stronger boost). Pass w into `partial_fit(`X`, `y`, sample_weight=`w`)`.

Role. Raises the CE share of novel / high-membership points without discarding known-class rows. Orthogonal to (1): same geometry, different row masses.

4 Cold-start gate ($n_c < 200$)

Let $n_{\text{idx}}(i)$ be the within-category arrival index of row i among novel labels. Define the cold-start mask

$$\mathcal{C} = \{i :\; y_i \in \mathcal{C}_{\text{novel}},\; n_{\text{idx}}(i) < N_{\text{cap}}\}, \qquad N_{\text{cap}} = 200. \tag{8}$$

- **Sample-level:** boost only on $\mathcal{C}$; warm novel ($n_{\text{idx}} \geq N_{\text{cap}}$) and known stay at $w_i = 1$.
- **Feature-level:** apply (1) only while $\mathcal{C} \neq \emptyset$; once every novel category is warm, skip scaling (geometry = uniform).

Budget on the learning curve = novel samples *per category*. Past N_{cap} the gate freezes empowerment: further data enter CE at uniform mass. No EWMA.

5 Four schemes

Scheme	Feature axis	Sample axis
`uniform`	raw X	$w_i \equiv 1$
`feature_w`	$X \cdot w^{\text{feat}}$ (1)	$w_i \equiv 1$
`sample_w`	raw X	fused w (7)
`both`	$X \cdot w^{\text{feat}}$	fused w

Loss = CE (`log_loss`) SGD; no EWMA. KPI = overall accuracy + mean novel OVA-AUC on a *fixed full LIVE holdout*. Budget b = novel samples / category, $b \in \{20, 40, \ldots, 200, 250, 300, 400\}$; known context $\approx 2 \cdot |\text{novel}|$.

6 Why two levers + the gate

- **Feature-level** answers: *which coordinates* carry the domain change? Stretch those axes.
- **Sample-level** answers: *which observations* look like the new category? Give them more CE mass.
- **Gate** answers: *how long* to empower? Only while $n_c < 200$.

7 FULL-holdout results (text; 3 seeds; gate@200)

Early cold-start ($b = 20$/cat). Mean overall acc: uniform 0.456, feature 0.461, sample 0.468, **both** 0.483 (Δ+0.027) — empowerment is visible when the new category is scarce.

At the cap ($b = 200$/cat). Mean Δacc vs U is near zero (gate about to freeze); `cnae9` still likes feature (+0.027).

Past the cap (endpoint $b = 400$/cat).

Dataset	uniform	feature_w	sample_w	both
news20_comp	0.502	0.499	0.504	**0.510**
news20_rec	0.545	0.550	0.550	**0.570**
news20_mix	0.504	0.507	0.502	**0.510**
cnae9_text	0.894	**0.916**	0.889	0.892
amazon_cat	0.737	**0.751**	0.747	0.744
mean overall acc	0.636	**0.645**	0.638	**0.645**
mean Δacc vs U	—	+0.009	+0.002	+0.009
mean novel OVA-AUC	0.840	0.838	0.836	0.833

Past 200, weighted schemes do *not* keep over-boosting: they sit slightly above uniform on overall acc; OVA-AUC stays flat. The gate does its job — empower the new category, then stop.

Reproduce.

```
python3 scripts/run_text_full_weight_curves.py --multiseed
```

Figures: `results/text_full_weight_curves/figures/`.